

ASSIGNMENT 1

Introduction to Web Development

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend Development:

Frontend development is the part of web development that deals with everything the user can see and interact with on a website. It includes the design, layout, buttons, forms, menus, animations, and other user-interface elements.

Common frontend technologies are **HTML, CSS, and JavaScript**.

Example: On an online shopping website, the product page, product images, search bar, buttons, navigation menu, and shopping cart interface are part of the frontend.

Backend Development:

Backend development deals with the server-side operations of a website. It manages business logic, authentication, databases, APIs, and requests received from the frontend.

Common backend technologies include **Node.js, Python, Java, PHP, and frameworks such as Express, Django, and Spring Boot**.

Example: When a user logs into an online shopping website, the backend verifies the username and password and retrieves the user's account information from the database.

Full-Stack Development:

Full-stack development involves both frontend and backend development. A full-stack developer can work on the user interface as well as the server, database, and APIs.

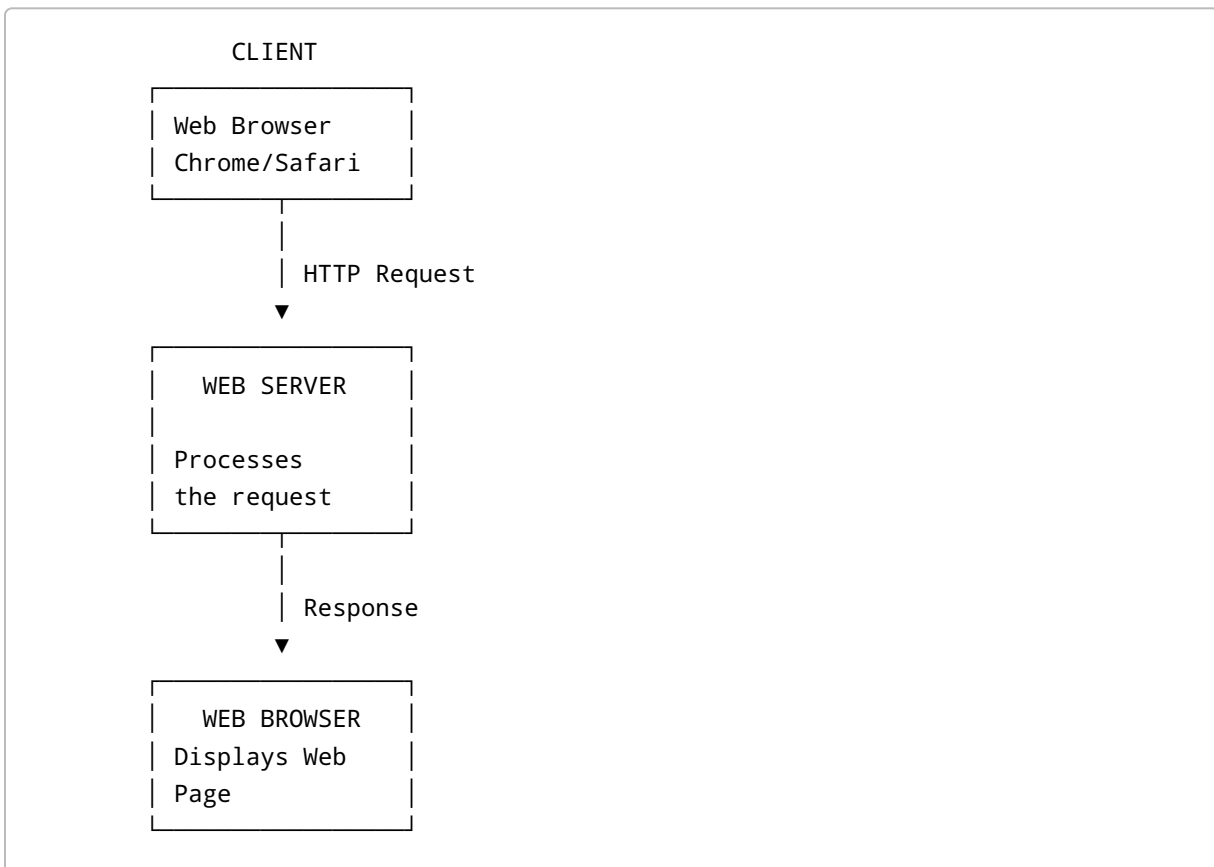
Example: A full-stack developer can build an entire food-delivery application, including the user interface, login system, restaurant database, order processing, payment integration, and APIs.

Feature	Frontend	Backend	Full-Stack
Main role	User interface	Server-side logic	Frontend + Backend
Runs mainly on	Browser	Server	Browser + Server
Technologies	HTML, CSS, JavaScript	Node.js, Python, Java, PHP	Combination of both
Example	Website interface	Login/API/database	Complete web application

2. Create a simple diagram showing how the client-server model works in web architecture.

The **client-server model** is a system in which a client sends a request to a server, and the server processes the request and sends a response back.

Diagram:



Working:

1. The user enters a website URL in a browser.
2. The browser sends a request to the web server.
3. The server receives and processes the request.
4. The server may communicate with a database if required.
5. The server sends a response back to the browser.
6. The browser displays the requested webpage to the user.

For example, when a user visits an online shopping website, the browser sends a request to the website server, which retrieves the required information and returns it to the browser.

3. Describe how a browser requests and displays a web page from a web server.

When a user enters a website address into a browser, several steps take place before the webpage appears.

Step 1: Entering the URL

The user enters a URL such as `https://example.com` into the browser.

Step 2: DNS Lookup

The browser uses the Domain Name System (DNS) to convert the domain name into the IP address of the web server.

Step 3: Establishing a Connection

The browser establishes a connection with the server using HTTP or HTTPS.

Step 4: Sending an HTTP Request

The browser sends an HTTP request to the web server asking for the required webpage.

Step 5: Server Processing

The server receives the request and processes it. If necessary, it may communicate with a database or another API.

Step 6: Server Sends Response

The server sends an HTTP response containing resources such as HTML, CSS, JavaScript, images, and other data.

Step 7: Browser Rendering

The browser reads the HTML and creates the DOM (Document Object Model). It loads CSS for styling and JavaScript for functionality.

Step 8: Displaying the Page

Finally, the browser renders the webpage and displays it on the screen.

```
User
  ↓
Enter URL
  ↓
DNS Lookup
  ↓
Web Server
  ↓
HTTP/HTTPS Request
  ↓
Server Processing
  ↓
HTTP Response
  ↓
```

HTML + CSS + JavaScript

↓

Browser Rendering

↓

Web Page Displayed

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

A basic web development environment requires several tools.

Tool	Purpose
Web Browser	Used to open, test, and debug websites
VS Code	Code editor used to write HTML, CSS, JavaScript, and other code
HTML	Used to create the structure of webpages
CSS	Used to style and design webpages
JavaScript	Used to add functionality and interactivity
Git	Version-control system used to track changes in code
GitHub	Online platform for storing and collaborating on code repositories
Node.js	JavaScript runtime used for server-side development and development tools
npm	Package manager used to install JavaScript libraries and packages
Terminal/Command Prompt	Used to run commands and development tools
Web Server	Used to serve webpages and handle client requests

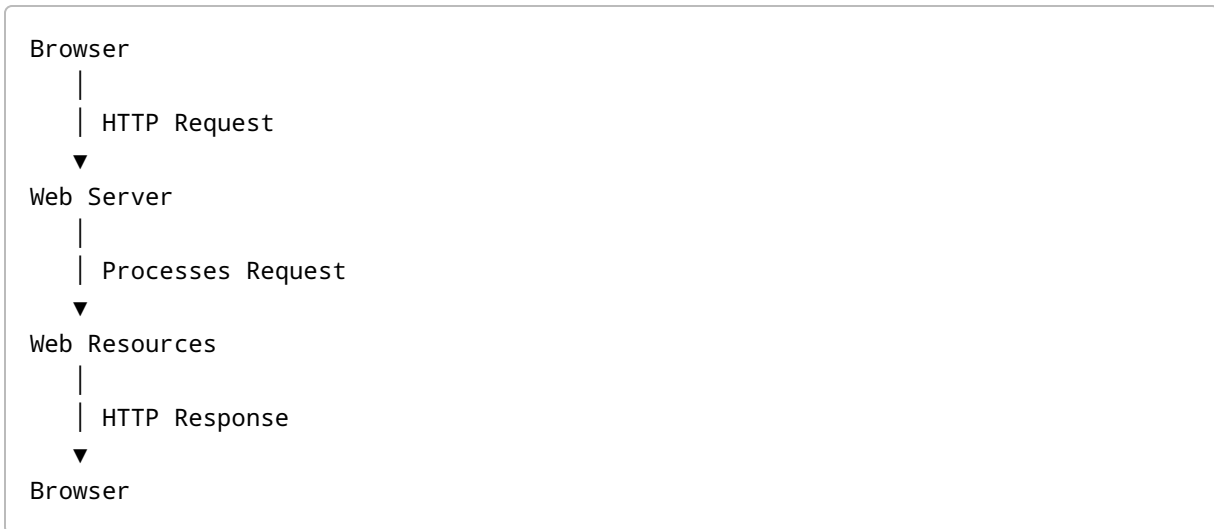
A beginner can start with **VS Code + Chrome + HTML + CSS + JavaScript + Git/GitHub**.

5. Explain what a web server is and give examples of commonly used servers.

A **web server** is software or a computer system that receives requests from clients, processes those requests, and sends web resources such as HTML pages, CSS files, JavaScript files, images, or other data back to the client.

The client is usually a web browser such as Chrome, Safari, Firefox, or Edge.

Working of a web server:



Commonly used web servers:

1. **Apache HTTP Server** – A widely used open-source web server.
2. **Nginx** – A high-performance web server and reverse proxy.
3. **Microsoft IIS** – Microsoft's web server for Windows environments.
4. **LiteSpeed** – A high-performance web server commonly used for hosting.
5. **Node.js HTTP Server** – Allows JavaScript applications to handle HTTP requests.

For example, when a user requests a webpage, the web server finds or generates the required content and sends it back to the user's browser.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Frontend Developer

A frontend developer is responsible for creating the part of a website that users see and interact with.

Responsibilities:

- Creating webpage structure using HTML.
- Designing webpages using CSS.
- Adding interactivity using JavaScript.
- Making websites responsive.
- Connecting the frontend to backend APIs.
- Testing the user interface.

Backend Developer

A backend developer works on the server-side portion of an application.

Responsibilities:

- Developing server-side logic.
- Creating and managing APIs.
- Implementing authentication and authorization.
- Processing user requests.
- Connecting applications to databases.
- Implementing business logic.
- Maintaining server-side security.

Database Administrator (DBA)

A database administrator manages the databases used by an application.

Responsibilities:

- Creating and maintaining databases.
- Managing database users and permissions.
- Backing up and recovering data.
- Monitoring database performance.
- Maintaining data security.
- Ensuring data consistency and availability.

Example:

In an online shopping application:

```
graph TD; A[Frontend Developer] --> B[Website Interface]; B --> C[Backend Developer]; C --> D[Business Logic / APIs]; D --> E[Database Administrator]; E --> F[Customer & Product Database];
```

All three roles work together to build and maintain a complete web application.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Visual Studio Code (VS Code) is a popular source-code editor used for web development.

Steps to install and configure VS Code:

1. Download and install **Visual Studio Code**.
2. Open VS Code.
3. Create a folder for the web development project.
4. Open the folder in VS Code.
5. Create a file named `index.html`.
6. Create a file named `style.css`.
7. Create a file named `script.js`.
8. Write HTML code in `index.html`.
9. Write CSS code in `style.css`.
10. Write JavaScript code in `script.js`.
11. Install useful extensions such as:
 12. Live Server
 13. Prettier
 14. HTML CSS Support
 15. JavaScript/TypeScript support
16. Open the HTML file using Live Server or directly in a web browser.

Basic project structure:

```
web-project/  
|  
├─ index.html  
├─ style.css  
└─ script.js
```

Example HTML:

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>My Web Page</title>  
    <link rel="stylesheet" href="style.css">  
  </head>  
  <body>  
  
    <h1>Hello World</h1>  
  
    <script src="script.js"></script>  
  </body>  
</html>
```

Screenshot:

After completing the setup, take a screenshot showing the **VS Code window, project files, and HTML code**, and attach it below this answer in the assignment.

8. Explain the difference between static and dynamic websites. Provide an example of each.

A **static website** displays fixed content. The content generally remains the same unless the developer manually changes the files.

A **dynamic website** generates or changes content based on user interaction, database information, login status, time, or other conditions.

Feature	Static Website	Dynamic Website
Content	Mostly fixed	Can change dynamically
Database	Usually not required	Commonly used
Complexity	Simple	More complex
Speed	Generally fast	Depends on processing
Technologies	HTML, CSS	HTML, CSS, JS + backend/database
Example	Portfolio website	Amazon, Instagram

Example of a static website:

A personal portfolio containing fixed information such as About Me, Skills, Projects, and Contact details.

Example of a dynamic website:

An online shopping website such as Amazon. Product information, prices, recommendations, shopping carts, and user accounts are dynamically generated and updated.

Simple representation:

STATIC:

Browser → Web Server → Fixed HTML → Browser

DYNAMIC:

Browser → Web Server → Backend → Database

↓

Dynamic Data

↓
Browser

9. Research and list five web browsers. Explain how rendering engines differ between them.

A **web browser** is software that allows users to access and interact with websites on the internet.

Five commonly used web browsers are:

1. **Google Chrome**
2. **Mozilla Firefox**
3. **Apple Safari**
4. **Microsoft Edge**
5. **Opera**

Browsers use **rendering engines** to convert HTML, CSS, and other web resources into the visual webpage displayed on the screen.

Browser	Main Rendering Engine
Google Chrome	Blink
Microsoft Edge	Blink
Opera	Blink
Mozilla Firefox	Gecko
Apple Safari	WebKit

Rendering Engines

Blink:

Blink is used by Chromium-based browsers such as Chrome, Edge, and Opera. It is responsible for processing HTML and CSS and rendering webpages.

Gecko:

Gecko is the rendering engine developed by Mozilla and used by Firefox.

WebKit:

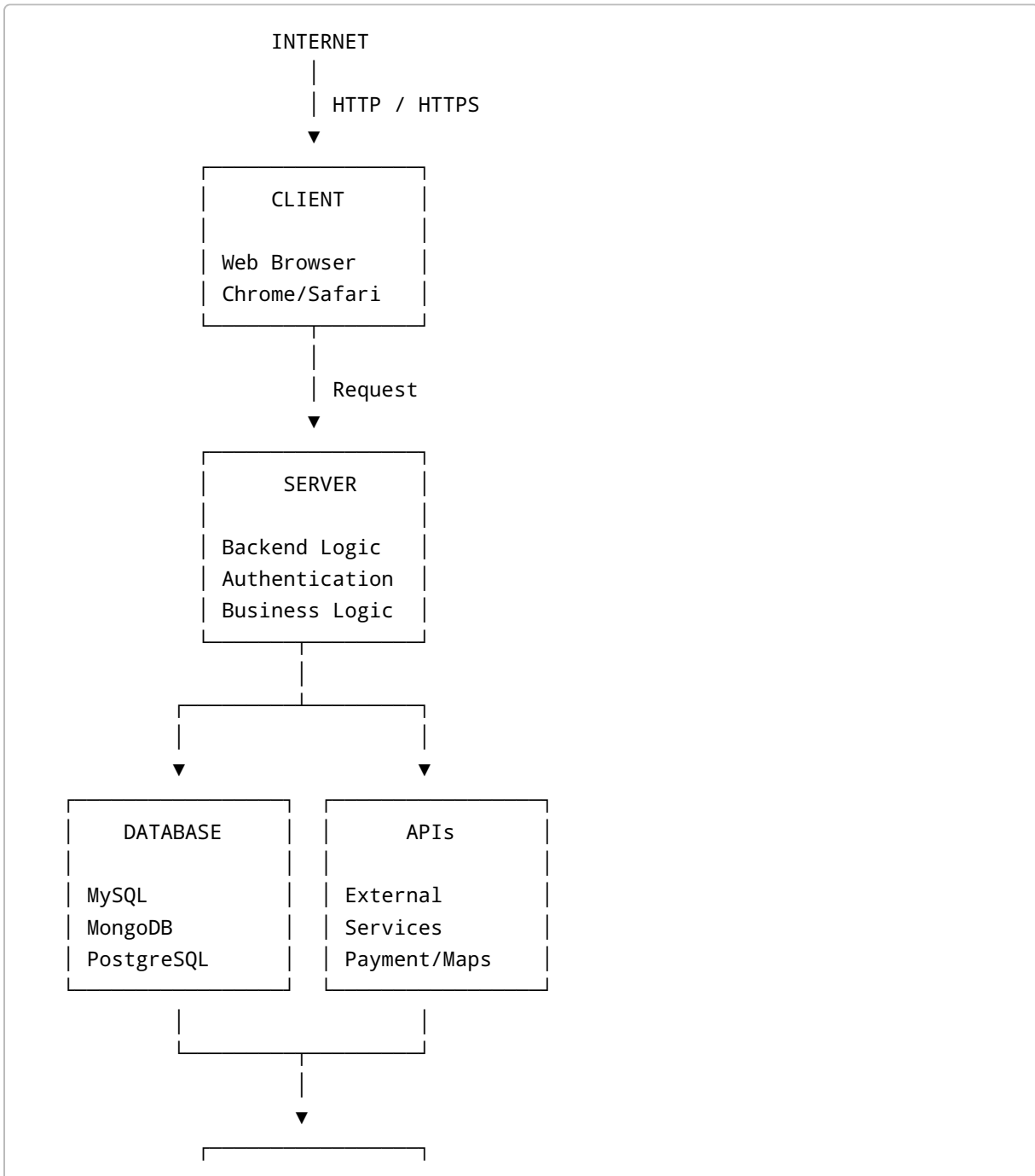
WebKit is Apple's browser engine and is used by Safari.

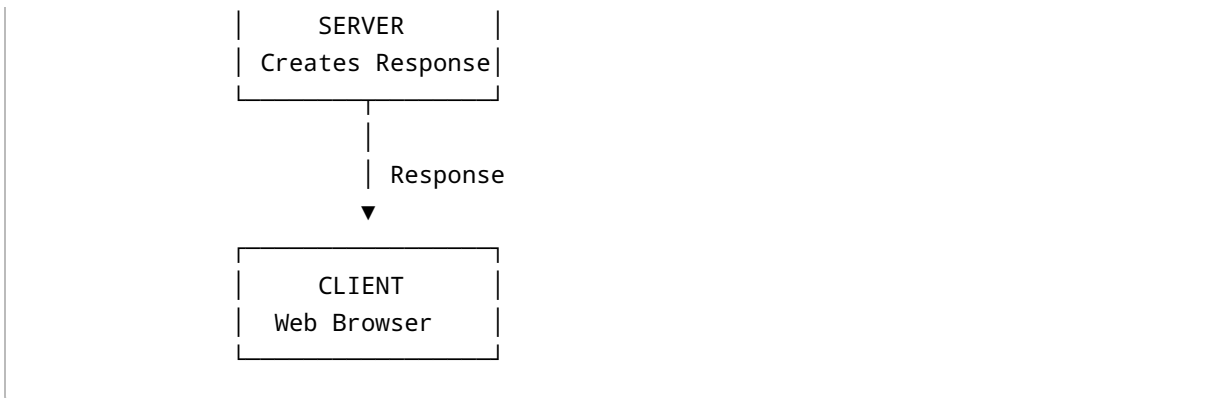
Rendering engines can differ in how they implement web standards, CSS features, JavaScript behavior, performance optimizations, and graphics rendering. Therefore, a website should be tested in multiple browsers to ensure compatibility.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

The basic web architecture consists of a **client**, **server**, **database**, and **APIs**. These components work together to provide information and services to users.

Labeled Diagram:





Explanation of the Architecture

1. Client:

The client is usually a web browser. It sends requests to the server and displays the response to the user.

2. Server:

The server receives client requests and performs the required processing. It contains backend logic and communicates with databases and APIs.

3. Database:

The database stores application data such as user accounts, products, orders, messages, and other information.

4. API:

An API (Application Programming Interface) allows different software systems to communicate with each other. A server can use external APIs to obtain services or information such as maps, weather, payments, or authentication.

Overall flow:

```
Client
↓
Request
↓
Server
↓
API / Database
↓
Server
↓
Response
↓
Client
```

↓
Webpage displayed to user

Thus, the client, server, database, and APIs work together to form the basic architecture of a modern web application.